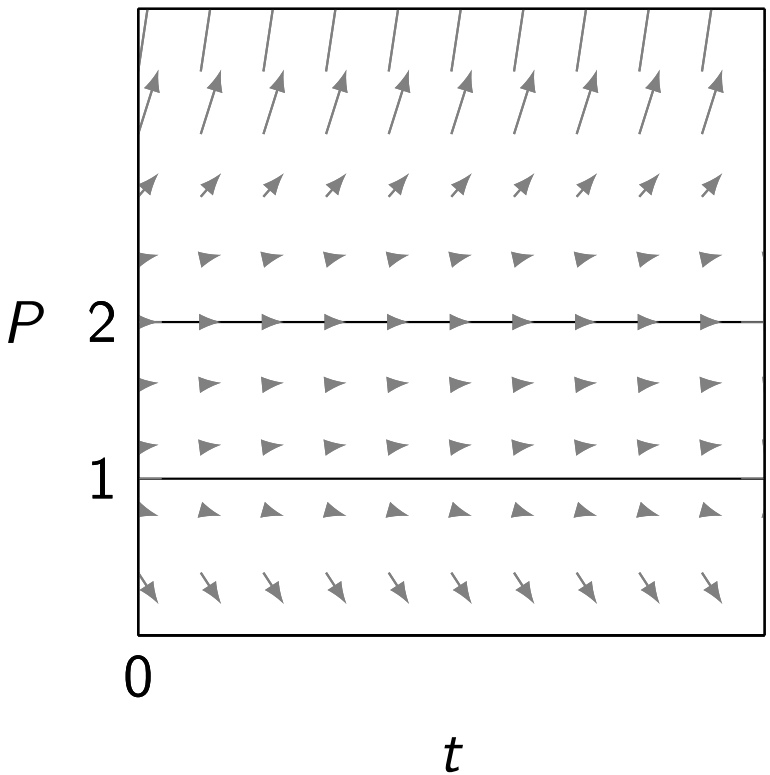
Lecture 09, Math 441

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Autonomous Equations and Slope Fields (cont.)

Ex: $\frac{dP}{dt} = (P-1)(P-2)^2$



Equilibrium solutions at

$P = 1$ (stable) and $P = 2$

(semi-stable).

$$\frac{dP}{dt} = \begin{cases} \textit{negative} & P < 1 \\ \textit{positive} & 1 < P < 2 \\ \textit{positive} & P > 2 \end{cases}$$

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The Logistic Equation with Harvesting

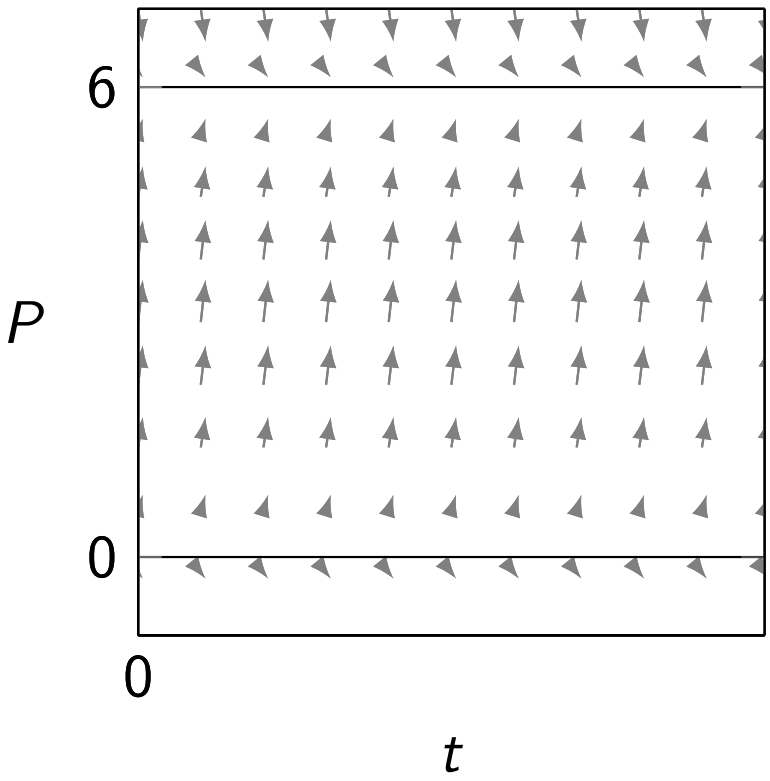
Ex: $\frac{\mathrm{d}P}{\mathrm{d}t} = 6P - P^2 - H$

H represents a constant harvesting term, independent of time and population. If for instance $\frac{\mathrm{d}P}{\mathrm{d}t}$ represented the growth of fish population, H would be fixed number of fish taken from that population (i.e. harvested) per unit time.

Let's consider different values of H.

$H = 0$

Ex: $\frac{\mathrm{d}P}{\mathrm{d}t} = 6P - P^2 = (6 - P)P$



Equilibrium solutions at

$P = 0$ (unstable) and $P = 6$

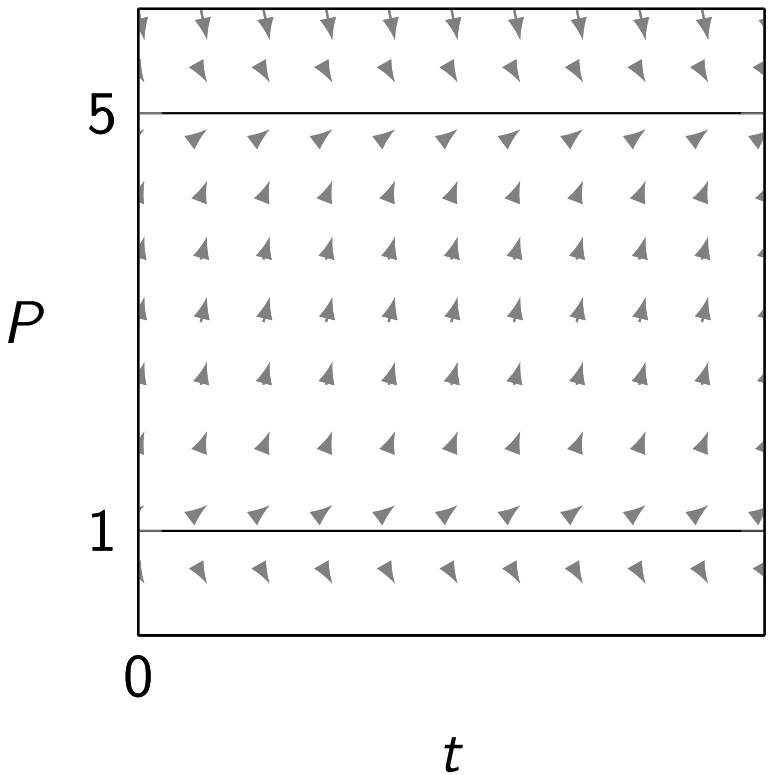
(stable).

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \begin{cases} \textit{negative} & P < 0 \\ \textit{positive} & 0 < P < 6 \\ \textit{negative} & P > 6 \end{cases}$$

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$H = 5$

Ex: $\frac{\mathrm{d}P}{\mathrm{d}t} = 6P - P^2 - 5 = -(P-1)(P-5)$



Equilibrium solutions at

$P = 1$ (unstable) and $P = 5$

(stable).

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \begin{cases} \textit{negative} & P < 1 \\ \textit{positive} & 1 < P < 5 \\ \textit{negative} & P > 5 \end{cases}$$

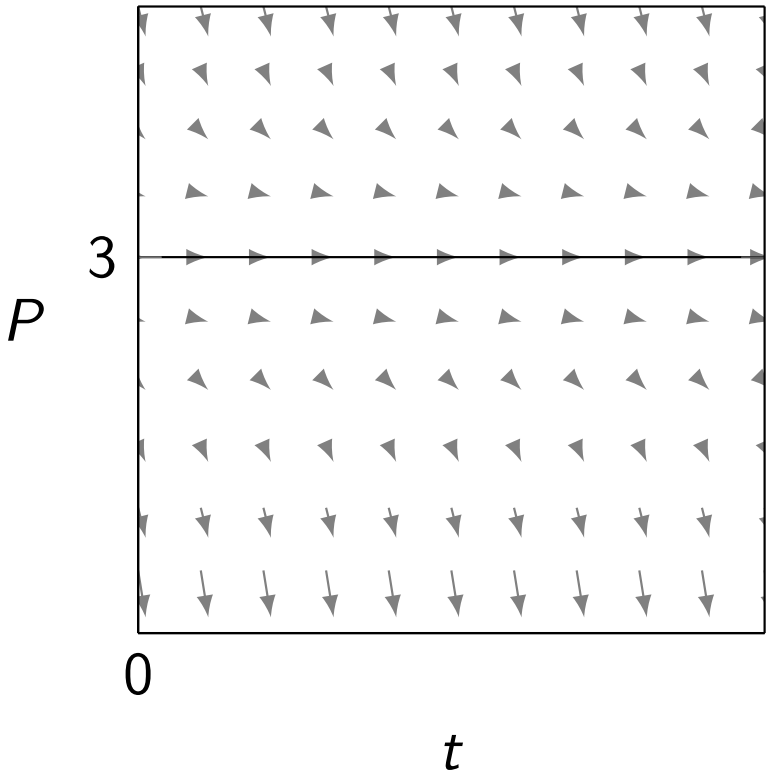
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The equilibrium has been lowered more because we are harvesting more. If the initial population is $P < 3$, the population will be extinguished.

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$H = 9$

Ex: $\frac{\mathrm{d}P}{\mathrm{d}t} = 6P - P^2 - 9 = -(P-3)^2$



Equilibrium solution only

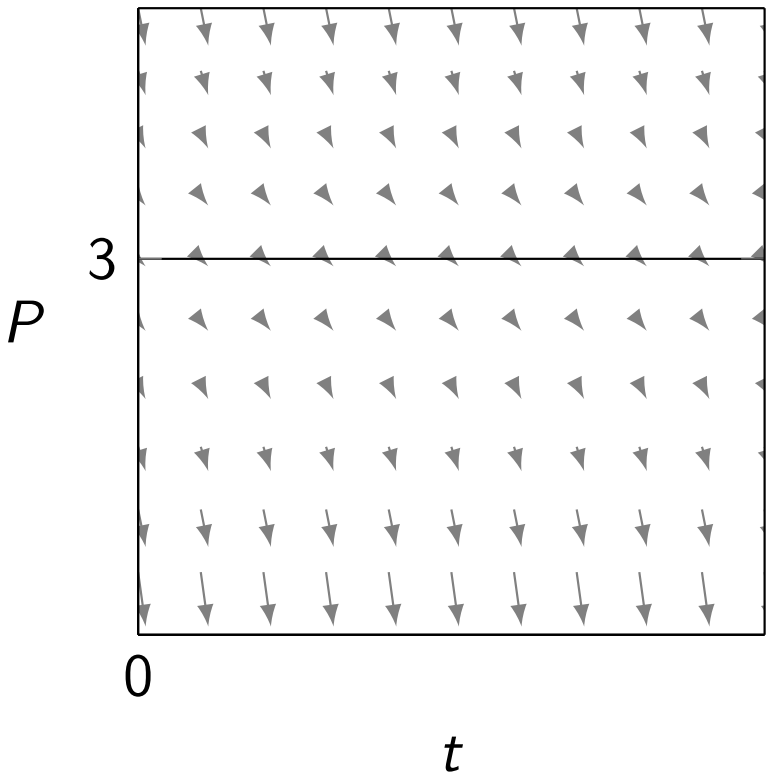
at $P = 3$ (semi-stable).

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \begin{cases} \textit{negative} & P < 3 \\ \textit{negative} & P > 3 \end{cases}$$

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$H = 10$

Ex: $\frac{\mathrm{d}P}{\mathrm{d}t} = 6P - P^2 - 10 = -(P-3)^2 - 1$



We're harvesting too much: no equilibrium solution! There is no value of P where the right side of the equation is zero. $\frac{\mathrm{d}P}{\mathrm{d}t}$ is negative everywhere.

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We can of course calculate the position of the equilibrium solutions as a function of H. Set the right hand side of the equation to 0: $6P - P^2 - H = 0$ and get

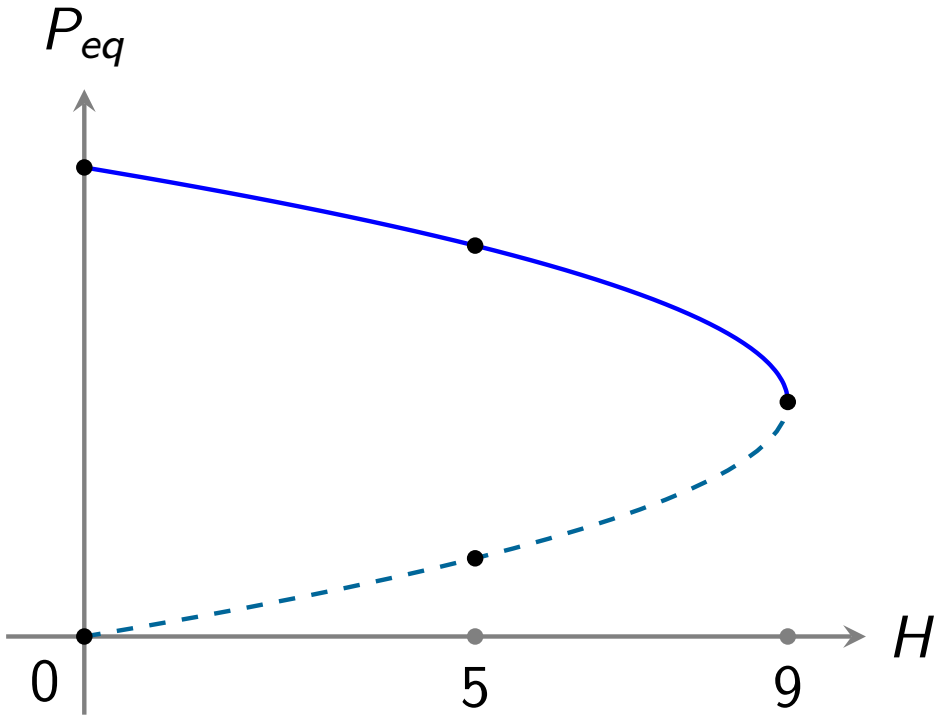
$$P_{eq} = 3 \pm \sqrt{9 - H}$$

which shows that there are equilibrium solutions only if $H \leq 9$

We can plot these curves on a bifurcation diagram.

Bifurcation Diagram

We can plot these curves on a bifurcation diagram that shows how the equilibrium solutions of the system change as a function of a parameter, H, and how there is a bifurcation at $H = 9$.



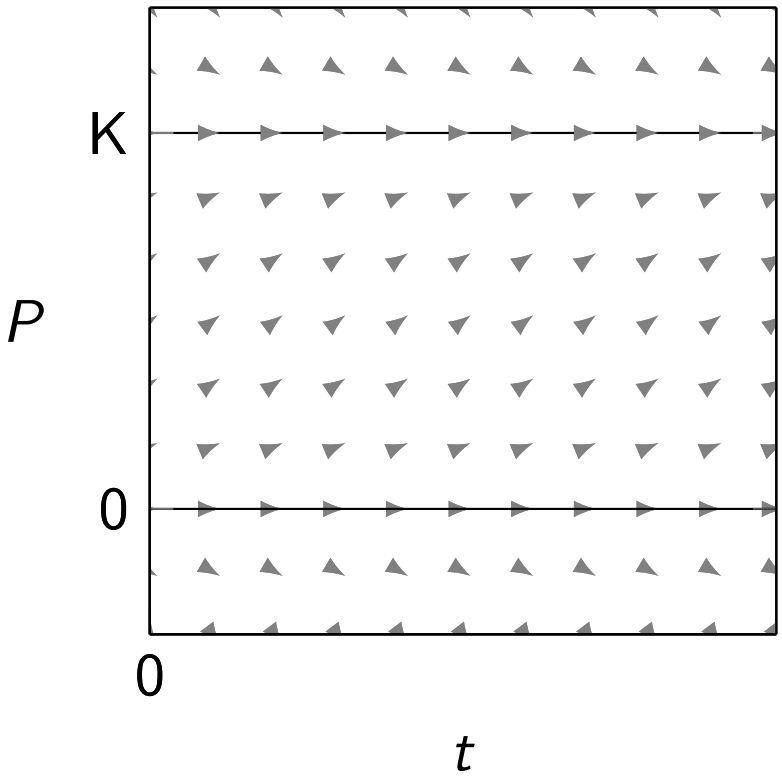
Solid lines are stable equilibrium solutions, and dashed lines are unstable.

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Harvesting Dependent on Population

The logistic equation without harvesting can be written as

$$\frac{\mathrm{d}P}{\mathrm{d}t} = R\left(1 - \frac{P}{K}\right)P$$



Equilibrium solutions at

$P = 0$ (unstable) and

$P = K$ (stable).

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \begin{cases} \textit{negative} & P < 0 \\ \textit{positive} & 0 < P < K \\ \textit{negative} & P > K \end{cases}$$

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Let's now add a harvesting term that depends on the population.

$$\frac{\mathrm{d}P}{\mathrm{d}t} = R\left(1 - \frac{P}{K}\right)P - E\,P, \qquad E \in [0, R]$$

E stands for effort. The idea is that we put a fixed amount of effort and we harvest more if the population is larger (easier to catch fish when there's more of them!), and less if P is small.

This is called a Schaefer model.

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Find the equilibrium solutions by setting $R\left(1-\frac{P}{K}\right)P - E\,P = 0$.

This has the solutions $P = 0$ (unstable), and $P_s = \frac{K}{R}(R - E)$

(stable). P_s is the sustainable population value (doesn't go to 0)

for a given value of E.

Let's calculate the max yield at the sustainable population:

$Y = E\,P_s$. This is how many fish we are getting, and using P_s:

$Y = \frac{K}{R}(R - E)\,E$

So the maximum yield is when $E = \frac{R}{2}$

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